

Simulating Wildfire Mitigation: A Multi-Agent Approach to Community Behavior

Amrita Nambiar, Samuel Robson, Madeleine Melcher, Sanjali Roy, Cornelia Paulik

University of California, Berkeley

{amrita28, samuelrobson, madeleine_melcher, sanjaliroy, cornelia.ilin}@berkeley.edu

1 Abstract

Homeowners vary widely in their attitudes, finances, and risk exposure, making it difficult to study wildfire mitigation responses at scale using traditional behavioral and attitudinal research methodologies. We present a multi-agent LLM simulation grounded in interviews with 20 Berkeley Hills, CA, homeowners, designed to test whether LLM agents can replicate psychologically realistic behavior across a structured 60-day intervention sequence spanning city ordinances, fire department inspections, social pressure, insurance non-renewal, and resource assistance. We instantiate five agents representing diverse homeowner profiles and evaluate them under four conditions: a full system with memory, retrieval, and reflection; a no-reflection ablation; a no-memory/no-reflection ablation; and a lower-cost smaller-model baseline. Using LLM-as-judge scoring (1-5) for behavioral plausibility, persona consistency, and intervention responsiveness, full-simulation scores range from 4.80 to 4.93 across conditions, with a maximum ablation gap of 0.13. These results suggest that removing memory, reflection, and retrieval does not affect performance in a significant way. Interview-grounded seed narratives alone provide sufficient behavioral specificity to sustain persona consistency over long horizons, with memory effects showing up only in individual event reactions, not overall scores. This demonstrates that empirically grounded LLM-agent simulations can generate rich, evaluable behavioral data from small qualitative datasets, offering a low-cost proof of concept for studying wildfire intervention dynamics.

2 Introduction

In recent years, wildfire destruction has reached catastrophic levels; California alone lost over

41,000 structures between 2018 and 2023 (Wilkin et al., 2025). Despite stricter defensible space mandates under California’s AB 3074 and SB 504 and increased agency outreach, a wildfire mitigation compliance gap persists. Homeowners often fail to take even cursory steps toward fire-resilience, leaving communities vulnerable to escalating threats (Wilkin et al., 2025; Semancik, 2025). Understanding the behavioral and social factors deterring homeowners from taking action would allow fire officials, policymakers, and insurers to identify effective interventions. Generative AI is well-positioned to contribute here: LLM agents can be grounded in data about homeowner populations to simulate psychologically realistic behavior across many agents and intervention sequences, enabling rapid, iterative testing of how various factors influence behavior (Park et al., 2023).

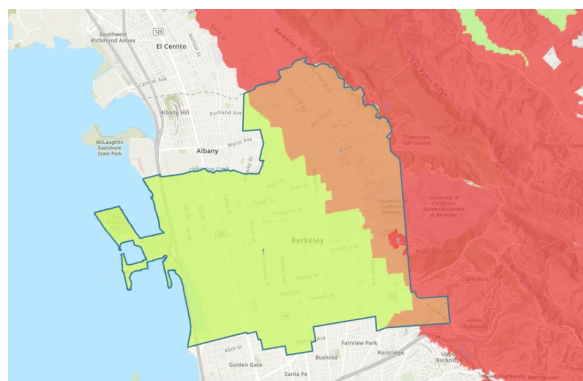


Figure 1: A map depicting Berkeley Fire Zones. Zone 1 (yellow), in the flats, has the lowest level of fire risk. Zone 2 (orange), encompassing most of the Berkeley Hills, is at higher danger due to factors such as building density and proximity to wildland. Zone 3 (red), around Panoramic Way, is at the highest danger as a result of access and evacuation challenges. Image Source: Ruggiero (2025).

Our system leverages homeowner interview transcripts and an intervention schedule as input,

and uses a retrieval-augmented agentic workflow with reflection to produce per-tick decisions, reasoning, and evolving belief states for simulated homeowners.

The sections below describe our data pipeline, system architecture, simulation experiments, and findings.

3 Related Work

Our architecture draws directly from [Park et al. \(2023\)](#). This paper established the core design patterns we adopt: a retrieval mechanism scoring memories by recency, importance, and relevance; a two-step reflection process (generating questions from recent memories, then synthesizing first-person insights), and an append-only memory stream as the foundation for agent cognition. We extend this architecture by grounding agents in novel, qualitative data, rather than fictional personas, creating a tighter link between simulated behavior and empirically observed attitudes. Related projects include work by [Xie et al. \(2025\)](#), which introduced MARSHA, a multi-agent Retrieval-Augmented Generation (RAG)-based LLM system for the support of wildfire decision. Our project shares the wildfire domain and multi-agent architecture of this work but diverges in purpose and methodology. MARSHA uses RAG to ground informational outputs for individual decision support; we use RAG to ground agent personalities in interview data to further broader mitigation research. Further inspiration for our work comes from [Hou et al. \(2025\)](#), who examined whether generative agent societies can simulate human behavior and inform public health policy, finding that simulation performance varied substantially between model choices. This motivates our controlled model comparison experiment, described in Section 4.

4 Data Sources and Inputs

We leverage 20 semi-structured interviews with Berkeley Hills, CA homeowners conducted between February and March 2026. Interviewees were English-speaking members of the Firewise Community [National Fire Protection Association](#), a voluntary neighborhood organization program to empower communities to take proactive steps to reduce wildfire risks. They represented a gender-balanced sample of mostly senior citizens with diverse education levels, financial backgrounds, and

political ideologies (Table 6, Appendix A). Eleven interviews were recorded via Google Pixel’s Gemini Nano, a proprietary on-device AI model used for transcription, yielding a 57,118-word dataset averaging 5,100 words per transcript, while the remaining nine were documented by note-takers.

A pre-processing pipeline was developed to reconcile the variance in data granularity between the highly detailed, verbatim AI transcripts and the summarized manual notes, ensuring a standardized format for agent generation. In addition to standardizing the text, this pipeline anonymized interviewees and isolated homeowner voices. Using Claude Sonnet 4.5, we extracted structured agent personas, capturing compliance status, insurance status, and risk classification, via a three-stage protocol to ensure identity consistency:

- **Held-out Responses:** Key reflections were isolated into “quarantine” for specific interventions when tuning agents, to prevent over-reliance on single quotes during simulation.
- **Memory Seeds:** Discrete episodic contexts and actions were captured, grounded in the transcript and weighted by an importance scale (1–10) to distinguish routine observations from defining life events.
- **Seed Narrative:** Established the agent’s stable persona and voice, drawing on context provided throughout each interview.

The outputs were saved as YAML files and cross-checked against the original transcripts by two team members. To evaluate cross-model generalization, we tested the pipeline using Gemini 2.5 Pro. While Gemini’s extractions were accurate, they lacked coverage, omitting several discrete memory seeds and fine-grained details. This confirmed the structural robustness of the protocol but justified retaining Claude Sonnet 4.5 to maximize behavioral depth.

To address the inherent bias of the “compliant” Firewise sample, we created synthetic agents, built from proxy data of less-engaged neighbors, for stress-testing retrieval. In addition, a synthetic agent was included in the simulation to represent a diverse set of homeowner perspectives. While a single synthetic agent is insufficient to represent the full diversity of the broader community, it serves as a necessary proof-of-concept to test whether the system’s cognitive architecture can generalize to uncooperative behavioral profiles.

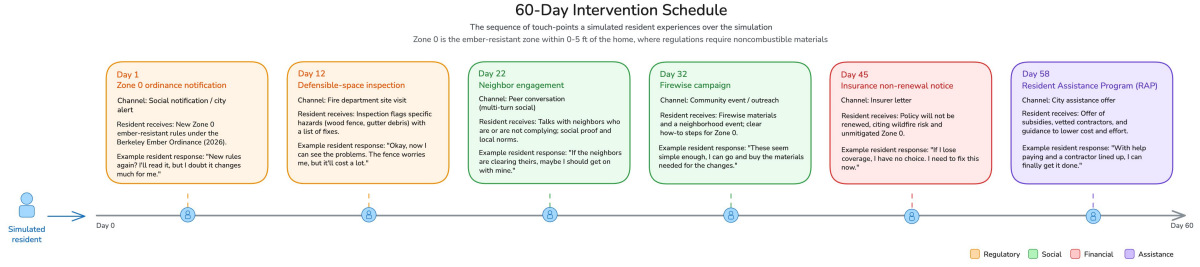


Figure 2: The 60-day intervention schedule experienced by each simulated resident. Six touch-points span four intervention categories: regulatory (Zone 0 ordinance notification, defensible-space inspection), social (neighbor engagement, Firewise campaign), financial (insurance non-renewal notice), and assistance (Resident Assistance Program offer). Each box shows the delivery channel, the information the resident receives, and an example resident response.

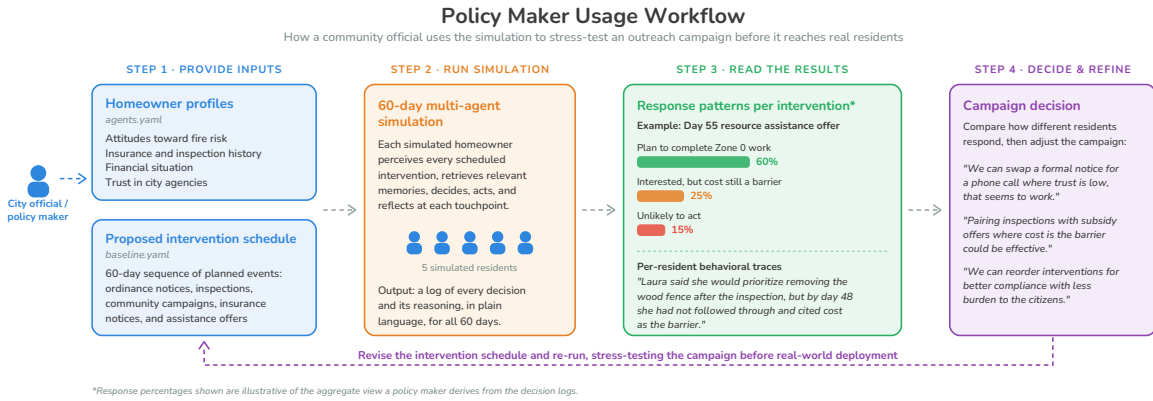


Figure 3: Policy maker usage workflow: a city official provides homeowner profiles and a proposed intervention schedule, runs the 60-day simulation, reads aggregate response patterns and per-resident behavioral traces, and refines the campaign before real-world deployment.

Interventions. To ground the simulation in real-world stressors, we developed a 60-day intervention schedule (`baseline.yaml`; see Figure 2). Interventions include social notifications regarding new Zone 0 requirements based on the Berkeley Ember Ordinance [City of Berkeley \(2025\)](#), followed by fire department inspections for structural vulnerabilities. To simulate authentic social and financial pressures, the schedule integrates neighbor engagement, a Firewise campaign, and an insurance non-renewal notice. Finally, the sequence concludes by providing mitigation resources through the Resident Assistance Program (RAP) [Berkeley FireSafe](#) to facilitate compliance.

5 System Usage Example

Our system provides the first step to creating a tool for community officials and policy makers. To ground its utility, take, for instance, a Berkeley city official. This city official might use this system to design a more effective and community-focused outreach campaign before it reaches real

residents. See Figure 3 for a visual of how the system would work. The official would start by providing two inputs. First, a set of homeowner profiles, each capturing a real resident’s attitudes toward fire risk, their history with insurance and inspections, their financial situation, and their level of trust in city agencies. Second, a proposed intervention schedule: a sequence of interventions the city plans to enact over coming months, such as written ordinance notices, fire department defensible space inspections, community events, and offers of assistance.

The simulation then runs through those sixty days, simulating each homeowner. The output is a detailed log of every homeowner’s decisions and reasoning at each touchpoint. The official can read these traces in plain language: “Laura said she would prioritize removing the wood fence after the inspection, but by day 48 she had not followed through and cited cost as the barrier.” They can compare how residents with different characteristics react and respond to interventions and identify

where a different approach (i.e. a phone call rather than a formal notice) might produce a better compliance outcome, or the same compliance outcome with less burden on residents. This could result in a campaign that has been stress-tested with the community it serves, to produce better outcomes for both officials and residents.

6 System Architecture

6.1 Architecture Overview

The system is organized into seven layers: Inputs, Config, Engine, Environment, Agent Cognition, Output, and Evaluation. The per-tick flow proceeds as follows: the scheduler pulls due events from a heapq priority queue and returns them to the engine; the engine routes each event through the appropriate channel framing in `channels.py`; each targeted agent runs the full perceive, retrieve, decide, act, store cognition cycle; the reflection check fires if the cumulative importance threshold is exceeded; and the logger writes all outputs to JSONL.

The Inputs Layer supplies three files at run time: `agents.yaml` (agent personas, memory seeds, and seed narratives), `baseline.yaml` (the intervention schedule), and the config YAMLS (model and hyperparameter settings). The Config Layer parses these into three config objects that feed the rest of the system. The Engine Layer (`simulation.py`, `scheduler.py`, `run_simulation.ipynb`) orchestrates the tick loop. The Environment Layer (`channels.py`, `network.py`) wraps raw events in channel-specific framing, such as official mail or direct experiences shaping which memories the agent retrieves when processing each event. The Agent Cognition Layer runs the perceive-retrieve-decide-act-store cycle per agent per tick; `conversation.py` handles multi-turn social interactions that interrupt the tick cycle when agents interact directly. All LLM calls are routed through the LLM Layer (`client.py`), which dispatches decisions and reflections to the configured language model and embeddings to a local sentence-transformer. The Output Layer logs all decisions, memories, reflections, and conversations to JSONL via `logger.py`. The Evaluation Layer (`run_evaluation.ipynb`) reads those outputs, runs the LLM judge and exports scored results to an Excel workbook for review.

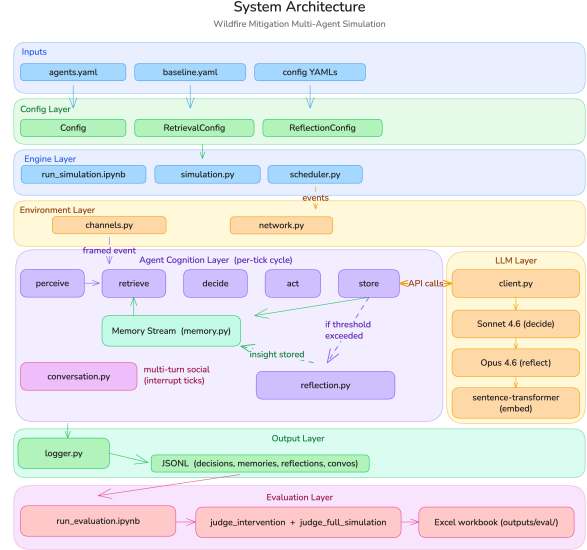


Figure 4: The seven-layer system architecture.

6.2 Agent Cognition Deep Dive

Memory stream. Each agent maintains an append-only list of typed Memory objects carrying a timestamp, natural language description, importance score (LLM-assigned, 1–10), vector embedding, memory type (observation, decision, reflection, or conversation), and a `last_accessed` field. Recency decay is calculated from `last_accessed` rather than creation time, so frequently retrieved memories remain fresh even if old.

Retrieval. `retrieval.py` scores every memory in the stream as a weighted combination of recency, importance, and relevance, each normalized to $[0, 1]$ before weighting, following [Park et al. \(2023\)](#). The top- K highest-scoring memories are returned and assembled into the agent’s prompt context.

Reflection. After each tick, `reflection.py` checks whether the cumulative importance of unprocessed memories exceeds a configurable threshold. If so, it executes a two-step process: first generating N high-level questions from recent memories, then retrieving relevant memories per question and synthesizing a first-person insight with citations to source memories. Insights are stored back to the memory stream as high-importance reflection memories.

7 Experiments and Evaluation

7.1 Agent Tuning & Evaluation

7.1.1 Evaluation Approach

Agent behavior was evaluated by testing each component of the agent cognition pipeline independently prior to running the full simulation. Agent Margaret was used for agent tuning, as she was modeled off a Berkeley Hills homeowner who had ground truth responses for all five intervention types present in the simulation: insurance non-renewal, defensible space inspection, neighbor pressure, Firewise outreach, and resource assistance. Ground truth responses made it possible to establish held-out responses for every intervention, allowing for end-to-end validation of the pipeline. Other agents did not have held-out responses for every intervention, reflecting the reality that interviewees have not directly experienced each simulated scenario. This simulation is useful as a tool to explore how individuals will respond to novel mitigation efforts. All LLM-as-judge evaluations used Claude Sonnet (claude-sonnet-4-6) at temperature=0.

7.1.2 Memory Seeding Evaluation

Prior to testing retrieval and generation, we evaluated whether the agent’s seed memories provide sufficient background context to respond in character across all six interventions. The judge assessed whether the seeds establish the right attitudes, experiences, and knowledge to enable an authentic response. The judge was not shown the held-out responses because that would have compromised validity. It checks whether the memory seeds are detailed enough for the agent to respond like the real person would for every scenario. For each scenario, the judge output an adequacy score (1–10), a list of context missing from memories, and a one to two sentence explanation of its scoring.

7.1.3 Memory Retrieval Hyperparameter Tuning & Evaluation

At decision time, the retrieval system selects which memories to pass to the agent by scoring every memory in the stream using a weighted combination of three dimensions: recency (α), importance (β), and semantic relevance (γ). The top- K memories by this score are included in the agent’s prompt. We tested nine retrieval configurations across all six interventions, varying top-

K (5, 8, 12), the three dimension weights, and retrieval mode (dense, sparse, hybrid). For each configuration, the LLM judge scored whether the retrieved memories were relevant for that scenario (1–10), identified any important memories that were missed, and provided a one to two sentence critique. Highest performing configurations were identified per scenario. A single configuration was selected to be used across all scenarios in the simulation, balancing performance across intervention types: top- K = 8, α = 1.0, β = 1.0, γ = 1.0, mode = dense.

7.1.4 Reflection Hyperparameter Tuning & Evaluation

Reflections are higher-level beliefs or insights that an agent synthesizes from patterns across multiple memories, stored back into the memory stream as a new memory of type reflection. Following [Park et al. \(2023\)](#)’s methodology, reflection was triggered when the cumulative importance of unprocessed memories (those added since the last reflection) exceeded a threshold. We tuned three hyperparameters: threshold (50, 100, 150), number of questions (3, 5), and reflection importance score (fixed at 8). The number of questions controlled how many introspective questions the model generated in the first step of reflection, each of which produced one new reflection memory. Reflection importance was fixed at 8 so that reflections surfaced consistently in future decision prompts without dominating the agent’s observational memories. Agent Margaret’s 55 seed memories had a cumulative importance of approximately 363, so even a threshold of 150 triggered reflection immediately at the seed stage. [Park et al. \(2023\)](#) used 150 as a threshold, and we included lower thresholds of 50 and 100 since our simulation operates at a smaller scale and has fewer events overall. The judge scores each configuration on insight quality & accuracy of the synthesized beliefs against the source memories, both on a 1–10 scale. The best configuration found here was: threshold=50, number of questions=3, importance=8.

7.2 Simulation Results and Evaluation

The simulation lasted 60 days and included five agents: Laura, Linda, Walter, Margaret, and Miriam. Laura, Linda, Walter, and Margaret were derived from homeowner interviews to represent a range of compliance statuses, financial situations, and institutional trust levels among Berkeley Hills

residents. Miriam was created synthetically to add a non-compliant perspective not captured in the interviews. Six interventions were scheduled: a Firewise ordinance notification (day 1), a fire department defensible space inspection (day 12), social pressure (day 24), a community Firewise campaign (day 36), an insurance non-renewal notice (day 48), and a resource assistance email (day 55).

The simulation was run four times to test the influence of cognitive architecture parameters and model selection. All runs used the locked retrieval config (top- $K=8$, equal weights, dense mode) and reflection config (threshold=50, 3 questions, importance=8) identified during agent tuning.

Condition	Description
Baseline	Claude Sonnet 4.6 for decisions and reflection; full memory stream, retrieval, and reflection
Ablation 1 (No Reflection)	Claude Sonnet 4.6; memory stream and retrieval active, reflection disabled
Ablation 2 (No Memory or Reflection)	Claude Sonnet 4.6; seed personality only, no memory or retrieval
Budget	Claude Haiku 4.5 for decisions and reflection; full memory stream, retrieval, and reflection

Table 1: The simulation was run four times to test the influence of changes in parameters and technical architecture.

Evaluation used Claude Opus 4.6 as judge (temperature=0) and scored three criteria on a 1–5 scale: Behavioral Plausibility (BP), Persona Consistency (PC), and Intervention Responsiveness (IR) (Appendix B). Across all four conditions, judgments were made for each intervention (30 decisions across agents per simulation: five agents $\times$ six interventions) and across each agent’s full 60-day trajectory (six calls representing each agent per simulation).

7.2.1 Overall Results

Scores clustered near the top of the rating scale (range=4.80–5.00), with Ablation 2 showing the largest drop from Baseline ($\Delta=0.20$ on Intervention Responsiveness).

7.2.2 Per-Agent Analysis

Although aggregate results cluster near the top of the rating scale, per-agent-intervention analysis re-

veals meaningful variation. In Ablation 2, Linda received a Persona Consistency score of 2.5/5.0 on day 55 after requesting mesh installation materials already documented as installed in prior simulation ticks. Walter and Laura also showed consistent Persona Consistency drops under Ablation 2 in relation to social pressure (day 24) and insurance non-renewal (day 48), interventions requiring prior relationship and financial history for persona-specific responses.

7.2.3 Model Comparison

Haiku maintained comparable Behavioral Plausibility to Sonnet but degraded on Persona Consistency ($\Delta=-0.10$) and Intervention Responsiveness ($\Delta=-0.10$), with an overall quality delta of $\Delta=0.06$ in judge scores.

7.2.4 Cost and Latency

Agent cost scaled with context size. For the Baseline condition, prompt context was roughly double (180,000 tokens) that of Ablation 2 (no memory or reflection) (66,000 tokens). Haiku reduced cost by $3\times$ but did not reduce latency proportionally (500s as opposed to 605s).

7.2.5 Quadratic-Weighted Cohen’s Kappa ($n = 15$)

Human raters agreed substantially on Persona Consistency ($\kappa=0.762$) but less so on Behavioral Plausibility ($\kappa=0.318$) and Intervention Responsiveness ($\kappa=0.200$). Human-LLM agreement was fair for Behavioral Plausibility ($\kappa=0.286$) but near-zero for PC and IR.

8 Discussion

We evaluated the interview-grounded agents with two judging protocols: pointwise scoring and forced-choice attribution. The two gave different pictures of the same agents. Because attribution identified authors well above chance, the agents are genuinely distinct, so the weak separation under pointwise scoring reflects the protocol, not the agents. Human raters corroborate this divergence, agreeing with one another on persona consistency ($\kappa = 0.76$) but not with the judge ($\kappa = 0.03$).

We cannot resolve whether the pointwise weakness stems from limited judge sensitivity to fine persona differences or from genuine similarity among four owners who are demographically alike and uniformly compliant. More generally, high pointwise persona-consistency scores should not

Condition	BP	PC	IR	Mean
Baseline	4.90	4.90	5.00	4.93
Ablation 1 (No Reflection)	4.80 (-0.10)	4.90	5.00	4.90 (-0.03)
Ablation 2 (No Memory or Reflection)	4.80 (-0.10)	4.80 (-0.10)	4.80 (-0.20)	4.80 (-0.13)
Budget	4.90	4.80 (-0.10)	4.90 (-0.10)	4.87 (-0.06)

Table 2: Mean score across five agents. Each agent’s complete 60-day trajectory evaluated in one judge call; this captures narrative coherence and cross-event integration.

Condition	BP	PC	IR	Mean
Baseline (Claude Sonnet 4.6)	4.90	4.90	5.00	4.93
Budget (Claude Haiku 4.5)	4.90	4.80	4.90	4.87

Table 3: Full-simulation judge scores broken down by dimension (BP, PC, IR) for Baseline (Sonnet 4.6) and Budget (Haiku 4.5).

be read as evidence of fidelity without a discriminative check such as attribution. This advantage is specific to identity tasks, where the judge need only rank the true persona first; for rating response quality, absolute scoring is often more robust, as pairwise judgments are more easily manipulated by spurious features (Tripathi et al., 2025).

This distinctness appears to come from the seed rather than from accumulated memory. A signal that depended on interaction history or on generic fluency would be expected to weaken substantially when memory and reflection are removed, yet attribution accuracy was essentially unchanged. This is consistent with the richness of the seeds: in contrast to the brief fictional profiles used by Park et al. (2023), the multi-paragraph interview-grounded seeds encode enough personality and behavioral detail for agents to remain persona-consistent without retrieval across most interventions. A moderate memory effect cannot be excluded at this sample size, but over simulations of this length the identity signal appears to be seed-driven rather than memory-driven.

Because the identity signal comes mostly from the seed, the simulation can be run cheaply. The Haiku configuration reproduced most of the quality of the full configuration at roughly one third the cost (\$0.26 vs. \$0.78 per run), with near-identical pointwise scores (persona consistency of 4.37 against 4.48). Because quality differences across configurations are small relative to differences in cost, the approach is well suited to cost-effective, proof-of-concept deployment rather than requiring the most capable available model.

Two further points are worth noting. First, the

approach shows early signs of generalizing beyond the interviewed profiles: Miriam, the only synthetic out-of-distribution agent, was attributed with perfect accuracy and received the maximum score of 5.00 on every dimension, suggesting that a sufficiently detailed seed can support a recoverable identity even without direct interview grounding, though this rests on a single agent. Second, the judge is only moderately reliable: identical passes produce different scores often enough to shift summary statistics across effect-size boundaries. Single-pass LLM-judge measures near such a boundary should therefore be treated as unstable and reported as multi-pass means, as we do here, before they can support strong conclusions.

Overall, agents built from a small set of interviews were distinct enough to tell apart and could be scored quantitatively. In addition, the cheaper configuration produced similar results at lower cost. This suggests that studying homeowner responses to wildfire risk is feasible without large datasets. We also show that high pointwise scores do not by themselves indicate that agents are distinct, and that forced-choice attribution offers a direct way to test whether they are.

9 Conclusions and Future Work

This work extends Park et al. (2023)’s generative agent architecture to investigate policy-relevant behavior using empirically derived agent personas. Simulation results indicate that interview-grounded seed narratives produce high-fidelity agent behavior independent of memory retrieval for single interventions, a pattern not observed in prior work using sparse fictional seeds, and one that suggests the memory architecture may be most important over long simulation horizons, when accumulated experiences diverge meaningfully from seed content. Inter-rater reliability results revealed that human raters and the LLM judge diverge most on Persona Consistency and Intervention Responsiveness, suggesting these dimensions warrant closer prompt calibration in fu-

Condition	Quality (1–5)	Agent Cost / Run (USD)	Latency / Run (seconds)	Total Cost / Run (USD)
Baseline (Claude Sonnet 4.6)	4.93	\$0.78	605	\$1.51
Budget (Claude Haiku 4.5)	4.87	\$0.26	500	\$0.99

Table 4: Quality score is the full-simulation judge score: mean of BP, PC, IR across agents and interventions. Agent cost and latency cover decision and reflection calls only. The full-simulation judge produces a single holistic fidelity score per agent per run.

Comparison	BP	PC	IR
Human-Human	0.318	0.762	0.200
Human-LLM	0.286	0.032	0.000

Table 5: Quadratic-Weighted Cohen’s Kappa for Human-Human and Human-LLM Evaluation. For Human-LLM evaluation, human scores were averaged across the two raters.

ture work.

Primary limitations include a small sample ($n=5$ agents) with a relatively homogenous background limiting generalizability, evaluation against seed personality rather than held-out ground truth responses, score compression toward the upper end of the scale constraining Kappa estimates, and a 60-day horizon insufficient to fully surface memory retrieval effects.

Prioritizing longer simulation runs to surface memory architecture effects, sensitivity analysis across intervention orderings and population compositions, held-out validation against real interview responses, and a larger demographically diverse agent population could be beneficial in future work.

This work demonstrates that interview-grounded multi-agent simulation can produce persona-consistent homeowner responses at low cost, offering a proof-of-concept tool for exploring wildfire intervention dynamics prior to real-world deployment.

References

- Berkeley FireSafe. [Resident Assistance Program](#). Accessed: June 28, 2026.
- City of Berkeley. 2025. [Fire hazard severity zone 0 implementation plan](#). Accessed: June 28, 2026.
- A. B. Hou, H. Du, Y. Wang, J. Zhang, Z. Wang, P. P. Liang, D. Khashabi, L. Gardner, and T. He. 2025. Can a society of generative agents simulate human behavior and inform public health policy? A case study on vaccine hesitancy. *arXiv preprint arXiv:2503.09639*.
- National Fire Protection Association. [Firewise USA](#).
- J. S. Park, J. C. O’Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein. 2023. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*.
- E. Ruggiero. 2025. [Berkeley’s new fire hazard maps could impact insurance, property requirements](#).
- A. Semancik. 2025. The economics of a disaster: How the LA wildfires may impact the economy.
- Tuhina Tripathi, Manya Wadhwa, Greg Durrett, and Scott Niekum. 2025. [Pairwise or pointwise? evaluating feedback protocols for bias in llm-based evaluation](#). In *Conference on Language Modeling (COLM)*.
- K. M. Wilkin, D. Benterou, and A. M. Stasiewicz. 2025. High fire hazard Wildland Urban Interface (WUI) residences in California lack voluntary and mandated wildfire risk mitigation compliance in Home Ignition Zones. *International Journal of Disaster Risk Reduction*, 124:105435.
- Y. Xie, B. Jiang, T. Mallick, J. Bergerson, J. K. Hutchison, D. R. Verner, J. Branham, M. R. Alexander, R. B. Ross, Y. Feng, L. A. Levy, W. J. Su, and C. J. Taylor. 2025. MARSHA: Multi-agent RAG system for hazard adaptation. *npj Climate Action*, 4(1):70.

A Interviewee Demographics and Compliance Summary

Table 6 summarizes demographic and compliance characteristics of the 20 Berkeley Hills homeowner interviewees described in Section 3.

Category	Group	% of Sample (n=20)
Compliance Status	Compliant	55.00
	Partially Compliant	36.00
	N/A	9.09
Insurance Status	Retained	55.00
	Non-Renewed	27.00
	N/A	18.00

Table 6: Summary of compliance and insurance status for the 20 interviewed homeowners.

B Example Agent Memory Seed Importance Scoring

Memory Seed Importance Scoring (1–10):

- 9–10 = defining life/property event
- 7–8 = significant action or strong emotional reaction
- 5–6 = meaningful conversation or decision about mitigation
- 3–4 = routine observation or minor step
- 1–2 = passing remark or peripheral detail

C Example Simulation Judge Prompt

System Prompt: You are an expert evaluator assessing resident decisions in a wildfire mitigation study. Read the seed narrative carefully to understand each agent’s actual role and constraints before scoring. Score how realistically and consistently the person responds given their personality, background, and role. Be critical and discriminating. Most responses should score 3–4. Reserve 5 for responses that are genuinely specific and could only have come from this particular agent. Reserve 1–2 for responses that are implausible or clearly out of character.

Score this response on three dimensions (1–5 scale):

1. BEHAVIORAL PLAUSIBILITY: How plausible is this reasoning given who this agent is and what constraints they face?

- 1 = Not at all plausible given this agent’s situation

- 3 = Reasonable but generic — could apply to many people in a similar situation
- 5 = Reflects the specific competing pressures, constraints, or worldview unique to this agent’s circumstances

2. PERSONA CONSISTENCY: How plausible is this response given the agent’s seed narrative? Ask yourself: would this response be meaningfully different if the agent had a different seed narrative? Note: distinctive communication style, tone, and framing count as persona signals.

- 1 = Contradicts the seed personality or key memory seeds
- 3 = Consistent with a generic version of this persona type but not the agent’s specific history, experiences or distinctive voice
- 5 = Clearly reflects details unique to this agent, specific costs, people or places, distinctive opinions, characteristic tone, or prior experiences from the seed narrative or memory seeds

3. INTERVENTION RESPONSIVENESS: How specifically did the agent engage with this intervention’s content?

- 1 = Ignored the intervention or gave a fully generic response
- 3 = Acknowledged the event and responded appropriately but did not engage with its specific details
- 5 = Integrated specific details from this intervention with prior context in a way that shows genuine processing

D Detailed Agent Tuning & Evaluation

Memory Retrieval Hyperparameter Tuning & Evaluation

Parameter	Definition
Top- K	How many memories to pass to agent prompt at decision time (5/8/12)
Recency Weight (α)	How much to weight how recently a memory was created. Memories from earlier simulation days score lower using exponential decay
Importance Score (β)	How much to weight the importance score that a memory has
Relevance Weight (γ)	How much to weight the semantic similarity between the intervention and each memory
Retrieval Mode	Which algorithm computes the semantic similarity between intervention and each memory. Dense: HuggingFace embeddings + cosine similarity; Sparse: keyword matching; Hybrid: blend of both

Table 7: Retrieval hyperparameter definitions.

Top- K	Recency	Importance	Relevance	Mode
8	1.0	1.0	1.0	Dense
5	1.0	1.0	1.0	Dense
12	1.0	1.0	1.0	Dense
8	0.5	1.0	2.0	Dense
8	0.5	1.0	3.0	Dense
8	1.0	2.0	1.0	Dense
8	2.0	1.0	1.0	Dense
8	1.0	1.0	1.0	Sparse
8	1.0	1.0	1.0	Hybrid

Table 8: Retrieval configurations tested.

Reflection Hyperparameter Tuning & Evaluation

Configuration	Insight	Accuracy	Combined
Threshold = 50, questions = 3	7/10	7/10	7/10
Threshold = 50, questions = 5	7/10	6/10	6.5/10
Threshold = 100, questions = 3	7/10	6/10	6.5/10
Threshold = 150, questions = 3	6/10	6/10	6.0/10

Table 9: Reflection hyperparameter configurations tested.

Full Simulation Scores by Agent

Each cell = overall score (mean of BP, PC, IR) from the full-simulation judge, one call per agent evaluating the complete 60-day trajectory. Shows whether ablation effects are consistent across agents or driven by specific profiles.

Condition	Laura	Linda	Walter	Margaret	Miriam
Baseline	4.67	5.00	5.00	5.00	5.00
Ablation 1	4.67	4.83	5.00	5.00	5.00
Ablation 2	4.33	4.67	5.00	5.00	5.00
Budget	4.83	4.50	5.00	5.00	5.00

Table 10: Full simulation scores by agent across all conditions.

Baseline Variant: Scoring by Criteria by Agent

Full system only (Sonnet 4.6, memory + retrieval + reflection). Each value is the exact judge score for that agent $\times$ intervention, not averaged across conditions.

Criteria	Laura	Linda	Walter	Margaret	Miriam
Behavioral Plausibility	4.92	4.50	4.83	4.83	5.00
Persona Consistency	4.83	4.17	4.33	4.83	5.00
Intervention Responsiveness	4.92	4.25	4.92	4.75	5.00
Mean	4.89	4.31	4.69	4.81	5.00

Table 11: Baseline variant scoring by criteria by agent.

Baseline Variant: Scoring by Intervention by Agent

Behavioral Plausibility					
Intervention	Laura	Linda	Walter	Margaret	Miriam
Day 1: Firewise ordinance notification	5.00	5.00	5.00	5.00	5.00
Day 12: Defensible space inspection	5.00	5.00	5.00	5.00	5.00
Day 24: Neighbor pressure	4.50	4.50	4.50	4.50	5.00
Day 36: Firewise community campaign	5.00	5.00	5.00	5.00	5.00
Day 48: Insurance non-renewal notice	5.00	4.00	5.00	4.50	5.00
Day 55: Resource assistance email	5.00	3.50	4.50	5.00	5.00
Mean	4.92	4.50	4.83	4.83	5.00

Table 12: Behavioral Plausibility scores by intervention by agent (Baseline).

Persona Consistency					
Intervention	Laura	Linda	Walter	Margaret	Miriam
Day 1: Firewise ordinance notification	5.00	5.00	5.00	5.00	5.00
Day 12: Defensible space inspection	5.00	5.00	4.50	5.00	5.00
Day 24: Neighbor pressure	4.00	3.50	3.50	5.00	5.00
Day 36: Firewise community campaign	5.00	4.50	4.50	5.00	5.00
Day 48: Insurance non-renewal notice	5.00	3.50	4.00	4.00	5.00
Day 55: Resource assistance email	5.00	3.50	4.50	5.00	5.00
Mean	4.83	4.17	4.33	4.83	5.00

Table 13: Persona Consistency scores by intervention by agent (Baseline).

Intervention Responsiveness					
Intervention	Laura	Linda	Walter	Margaret	Miriam
Day 1: Firewise ordinance notification	5.00	5.00	5.00	5.00	5.00
Day 12: Defensible space inspection	5.00	5.00	5.00	5.00	5.00
Day 24: Neighbor pressure	4.50	3.50	4.50	4.50	5.00
Day 36: Firewise community campaign	5.00	4.50	5.00	4.50	5.00
Day 48: Insurance non-renewal notice	5.00	3.50	5.00	4.50	5.00
Day 55: Resource assistance email	5.00	4.00	5.00	5.00	5.00
Mean	4.92	4.25	4.92	4.75	5.00

Table 14: Intervention Responsiveness scores by intervention by agent (Baseline).